EARTHPULSE

An interconnected ecological restoration intelligence system
Four-year engineering and research mission | 2026-2030

Measure. Understand. Act. Verify.

Build an open-source, distributed environmental intelligence platform that helps communities, researchers and restoration organizations detect ecological degradation, decide where intervention is useful and evaluate whether restoration works.

Key principle: Monitoring environmental conditions is not the same as restoring an ecosystem. EarthPulse is meant to provide credible evidence and decision support to restoration practitioners, not replace ecological expertise.

The long-term vision

Connect forests, rivers, soil, wildlife observations and satellite data through one resilient platform. Begin with a single accessible field site, low-cost instruments, standardized ecological measurements and a trustworthy data pipeline.

1. The environmental systems

Domain	Potential observations	Important limitation
Forest / vegetation	Native plant survival, canopy structure, seasonal vegetation patterns	Greenness is not equivalent to biodiversity.
Water	Temperature, turbidity, dissolved oxygen, pH and sampling records	Affordable sensors require calibration and maintenance.
Soil	Moisture, erosion surveys, laboratory tests	Cheap probes do not automatically yield accurate soil moisture percentages.
Biodiversity	Bird/frog sounds, expert-reviewed observations, habitat indicators	Sound detections are not a complete species census.

2. Target architecture

Field layer: ESP32/C++ sensor nodes, standardized vegetation and water surveys, passive acoustic recorders and relevant geospatial datasets.

Edge and communications: Sampling scheduler, local buffering, fault handling, power management, Wi-Fi initially and LoRa where deployment conditions justify it.

Data platform: Authenticated ingestion endpoints; FastAPI; PostgreSQL/PostGIS; time-series data; device identity, metadata and quality flags. MQTT can be introduced later for larger fleets.

Research and AI: Quality control, statistical analysis, satellite change detection, acoustic classification, anomaly detection and explicit uncertainty estimates.

Restoration decisions: Dashboards, mapped observations, evidence summaries and human-reviewed recommendations.

What each of your engineering domains contributes

Discipline	Technical focus
C++	Embedded firmware, sensor drivers, robust communications, memory efficiency, scheduling.
ENTC	Signal conditioning, power electronics, instrument design, calibration, DSP, wireless networks.
Backend	APIs, Postgres/PostGIS, permissions, observability, distributed ingestion, datasets.
AI / ML	Time-series methods, geospatial analysis, audio recognition, uncertainty, model evaluation.
Research	Baselines, comparison plots, repeated surveys, reference measurements and reproducible experiments.

3. The four-year roadmap

Year	Engineering work	Must-have deliverable
Year 1 2026-27	One ESP32 monitoring node; embedded C++; API, database and dashboard; site permission; reference measurements and initial ecology learning.	Field-tested prototype, documented firmware/backend, calibrated observations and baseline field report.
Year 2 2027-28	Add water/vegetation observation methods, public geospatial datasets, acoustic recordings and a first evaluated classifier. Use statistical baselines before ML.	Versioned multi-domain dataset, validation results and explainable ecological indicators.
Year 3 2028-29	Scale to ~5-10 nodes only if feasible. Develop offline buffering, LoRa where useful, deployment tooling, device fleet health and a comparison-based restoration study.	Reproducible deployment toolkit, reliability report and early intervention evaluation.
Year 4 2029-30	Partner with restoration practitioners; evaluate decision usefulness and ecological outcomes; publish reproducible software, hardware and research results.	Open-source EarthPulse V1, external field collaboration and a strong research/final-year project.

Reality check: A credible four-year outcome is a robust research prototype validated at one or more sites, not proven landscape-scale ecosystem recovery.

4. First working prototype: EarthPulse Node V1

Initial objective: Measure air temperature, relative humidity and raw soil-probe readings; transmit authenticated time-stamped data; visualize trends and missing readings; perform a safe field trial at an authorized location.

Proposed hardware and indicative budget (INR)

Part	Planning allowance
ESP32 development board	₹400-700
BME280 environmental sensor	₹250-650
Capacitive soil moisture probe	₹150-400
Breadboard, wiring and connectors	₹200-400
Weather-resistant enclosure and cable glands	₹400-1,000
USB power bank or safe regulated power supply	₹500-1,200
Miscellaneous and replacement parts	₹500-1,000
Total estimated allowance	**₹2,400-5,350**

Budget is illustrative, not a verified retail quote. Begin with Wi-Fi and regulated USB power. Add solar charging, battery protection or LoRa only when field requirements justify them.

Telemetry fields

device_id, **timestamp (UTC)**, air_temperature_c, relative_humidity_pct, soil_moisture_raw, firmware_version, optional battery_mv, calibration_version and explicit measurement-quality flags. Distinguish raw, calibrated, estimated and missing data.

Minimal HTTP API

POST /api/v1/telemetry

GET /api/v1/devices

GET /api/v1/devices/{id}/readings

GET /api/v1/sites/{id}/summary

GET /api/v1/sites/{id}/observations

POST /api/v1/observations

Use per-device authentication and HTTPS where supported. Keep the first backend simple; add MQTT only when it solves a real fleet requirement.

5. First 12 weeks: prototype execution

Weeks	Work package	Acceptance test
1-2	Choose authorized field site; identify ecology advisor; define research question and sampling protocol.	Site permission and baseline protocol recorded.
3-4	Assemble ESP32 sensors; implement C++ firmware; compare with reference instruments.	Measurements logged; calibration/limitations documented.
5-6	Build FastAPI ingestion; PostgreSQL storage; device auth; validation and data quality.	Valid records persist; invalid requests rejected.
7-8	Develop React dashboard; time-series plots; sensor availability; site observations.	Dashboard shows real measurements and missing data.
9-10	Outdoor field test; check enclosure, communications, drift, battery/power and failure behavior.	Deployable node with measured uptime/data delivery.
11-12	Analyze reference agreement; publish technical report, source, wiring and deployment notes.	Reproducible v0.1 prototype and limitations report.

6. Research agenda and field validation

Primary research question: Can low-cost distributed environmental sensing, combined with standardized field observations, supply reliable and actionable evidence for small-scale restoration decisions?

Question	Method	Metric
Are inexpensive sensors dependable?	Compare readings with reference instruments across field conditions.	Bias, mean absolute error, sensor drift.
Does the node function unattended?	Run a documented multiweek deployment.	Uptime, missing-data fraction, power use, maintenance.
Can unusual conditions be identified?	Start with statistical methods, then test anomaly models.	False alert rate, precision, recall.
Can sound support biodiversity monitoring?	Expert-label a sample, compare model detection with labels.	Per-species precision and recall.
Are interventions working?	Repeated native vegetation surveys, comparison areas and seasonal controls where feasible.	Survival rates, cover, habitat indicators and uncertainty.

For credible impact assessment, use reference and comparison sites and, where feasible, a before-after-control-impact experimental design. Avoid claims of causality without adequate controls and time coverage.

7. Open datasets and reference projects

Copernicus Data Space — Satellite imagery for change analysis
https://dataspace.copernicus.eu/

ISRO Bhuvan — Indian geospatial and thematic resources
https://bhuvan.nrsc.gov.in/

GBIF — Species occurrence datasets
https://www.gbif.org/

India Biodiversity Portal — Indian biodiversity and citizen science
https://indiabiodiversity.org/

Open Acoustic Devices — Acoustic monitoring hardware and resources
https://www.openacousticdevices.info/

Arbimon — Acoustic biodiversity monitoring
https://arbimon.org/

Society for Ecological Restoration — Restoration standards and research resources
https://www.ser.org/

Global Forest Watch — Forest monitoring applications
https://www.globalforestwatch.org/

Research caution: biodiversity observations can have spatial sampling bias; satellite greenness is not a proxy for ecological health on its own. Confirm licensing and use limitations before incorporating third-party datasets.

8. Weekly learning plan (12 project-focused hours)

Area	Hours/week	Application
C++ and electronics	4	Sensor drivers, embedded logic, circuits and debugging.
Backend and databases	3	Ingestion, data model, auth, dashboard APIs.
Math, statistics and ML	2	Uncertainty, signal analysis, evaluation.
Ecology and field research	2	Read studies, define protocols, validate indicators.
Documentation	1	Research journal, diagrams, decisions, test reports.

This is project-specific time, not a replacement for degree coursework or separate DSA practice. Prioritize C++ fundamentals, electronics, backend/database engineering, probability and statistics, then applied ML.

9. Proposed 2030 outcomes

- Approximately 5-10 monitoring nodes if funding, permissions and maintenance allow.
- At least two environmental measurement domains integrated with appropriate validation.
- At least one serious collaboration with an ecologist, university or restoration organization.
- An open-source reproducible platform including firmware, schema, API, field protocols and deployment documentation.

- A well-supported technical or research report, potentially suitable for publication if results merit it.

10. Risks, safeguards and non-goals

- **No AI theater:** do not build elaborate prediction systems without measured, useful field data.
- **No false restoration claims:** monitoring outcomes are not proof of ecological recovery or causality.
- **No hardware extravagance:** validate simple sensors and maintenance workflows before scaling.
- **Responsible fieldwork:** obtain permissions, respect wildlife, safeguard sensitive species locations and protect local community information.
- **Own footprint:** account for batteries, discarded electronics, transport and cloud costs.
- **Human expertise:** let qualified ecologists and local practitioners shape interventions.

11. Immediate next actions

- Create the **earthpulse** repository with **firmware/**, **backend/**, **dashboard/**, **field-protocols/** and **research/**.
- Find an accessible, authorized site and speak to an ecologist or local restoration group about their actual data needs.
- Build one ESP32 node and one authenticated ingestion endpoint before purchasing more hardware.
- Start research note **EarthPulse R001: Design and Validation of a Low-Cost Environmental Monitoring Node for Ecological Restoration**.

Final idea: Develop excellent, transferable engineering ability in service of a concrete environmental problem. A modest, repeatable system that helps one restoration team is more valuable than an impressive dashboard with unsupported claims.